



Full Length Article

Leap-SLAM: Degeneracy-mitigated robust SLAM with leapfrogging multi-robot collaboration in tunnels

Shoubin Chen^{a,b}, Jiasheng Chen^{a,b}, Baiyang Zhang^a, Maosheng Yan^{a,c}, Jianping Li^d,
Bo Zhang^{a,b,*}, Qingquan Li^{a,b}

^a Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ), Shenzhen, 518107, China

^b Shenzhen University, Shenzhen, 518060, China

^c State Key Laboratory of Information Engineering in Surveying Mapping and Remote Sensing (LIESMARS), Wuhan University, Wuhan, 430072, China

^d School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, 639798, Singapore, Singapore, 639956, Singapore

ARTICLE INFO

Keywords:

Cooperative localization
Laser mapping
Leapfrogging collaboration
Degeneracy mitigation
Tunnels
Robot inspection

ABSTRACT

With the rapid advancement of embodied robotics, robot inspection and detection in Global Navigation Satellite System (GNSS)-denied tunnels remains challenging. Light Detection and Ranging (LiDAR) based Simultaneous Localization and Mapping (SLAM) often suffers from degeneracy due to the scarcity of geometric features, leading to odometry drift and mapping inconsistencies. In this paper, we propose Leap-SLAM, a degeneracy-mitigated robust LiDAR-Inertial framework that effectively fuses high-localizability constraints via a leapfrogging multi-robot collaboration strategy. Specifically, we design degeneracy indices based on the information matrix for real-time uncertainty estimation. A Beacon Robot (BR) with reflective markers is introduced to provide external stable constraints, enabling a Master Robot (MR) to perform accurate state estimation even in feature-poor environments. Experimental validation in three diverse tunnels demonstrates superior performance over state-of-the-art methods. Notably, Leap-SLAM exhibits exceptional stability, maintaining high trajectory consistency and a negligible drift rate even after long-range traversals in feature-starved tunnels. This methodology offers a practical solution for robotic inspection in geometrically degraded engineering environments. The source code and datasets are publicly available on GitHub¹.

1. Introduction

Underground tunnels constitute critical civil infrastructure whose structural integrity directly underpins urban safety and functionality (e.g., transportation networks and utility services) [1–3]. In these Global Navigation Satellite System (GNSS)-denied and structurally complex environments, Simultaneous Localization and Mapping (SLAM) has traditionally served as the backbone for robotic state estimation. However, the emerging paradigm of Embodied Artificial Intelligence (AI) and Large Models demands a transition from passive geometric mapping to active, semantic-aware perception [4]. In this context, SLAM is no longer merely a localization tool but acts as a crucial fusion engine. It integrates multi-modal sensory data to provide a robust spatio-temporal anchor, enabling embodied agents to bridge low-level sensor streams with high-level cognitive reasoning for autonomous inspection¹.

Most existing SLAM technologies are single-robot methods [5–10] that excel in feature-rich environments. However, in geometrically degraded settings such as underground tunnels—where repetitive structures yield insufficient reliable constraints—these methods degrade substantially [11]. While degeneracy-aware approaches can analyze and partially mitigate estimation failure [12–15], single-robot systems remain fundamentally limited by intrinsic perception constraints and the absence of external absolute references.

Multi-robot SLAM [16–18] has emerged as a promising alternative, expanding the observation space to provide external constraints that enhance stability, and has demonstrated significant advantages in various application scenarios. However, most existing multi-robot frameworks primarily focus on decentralized optimization and multi-agent pose consistency [19], overlooking the potential of active collaboration to synthesize external stable constraints for feature-starved environments.

* Corresponding author.

E-mail addresses: shoubin.chen@whu.edu.cn (S. Chen), chenjiasheng@gml.ac.cn (J. Chen), zhangbaiyang@gml.ac.cn (B. Zhang), yanms_sgg@whu.edu.cn (M. Yan), jianping.li@ntu.edu.sg (J. Li), zhangbo@szu.edu.cn (B. Zhang), liqq@szu.edu.cn (Q. Li).

¹ <https://github.com/G-SEI-Lab/Leap-SLAM>

<https://doi.org/10.1016/j.inffus.2026.104538>

Received 19 January 2026; Received in revised form 14 May 2026; Accepted 5 June 2026

Available online 10 June 2026

1566-2535/© 2026 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Consequently, ensuring degeneracy-mitigated, stable localization without reliance on sufficient environmental inherent constraints or the complex fixed-mounted location beacons [20] remains an open challenge, necessitating novel external constraints based on multi-robot collaboration mechanisms.

We address this through Leap-SLAM, a degeneracy-mitigated robust LiDAR-Inertial framework that effectively fuses high-localizability constraints via a leapfrogging multi-robot collaboration strategy. The system comprises a Master Robot (MR) performing localization and mapping, and a Beacon Robot (BR) that carries portable reflective markers to provide reliable high-localizability constraints. Distinct from classical symmetric multi-robot SLAM that relies on redundant map merging and bidirectional information exchange, our approach introduces a multi-robot SLAM paradigm that achieves both collaborative localization and consistent 3D point cloud mapping. By utilizing the BR as an active mobile reference to provide deterministic geometric constraints for the MR, this strategy effectively mitigates observability degradation in feature-poor environments. This architecture enhances system efficiency and robustness by intentionally avoiding the computational overhead and communication redundancy inherent in traditional mapping fusion. The main contributions of the proposed work are as follows:

1. We propose a degeneracy-mitigated robust SLAM framework driven by the leapfrogging multi-robot collaboration strategy in degraded tunnels. This framework introduces a BR with reflective beacons and designs a leapfrogging cooperative advancement strategy for the MR and the BR. Crucially, the proposed method fuses the high-localizability feature information provided by the BR with local LiDAR-Inertial measurements. This multi-source fusion maintains continuous and reliable observations, thereby mitigating geometric degeneracy for accurate state estimation in degraded tunnels.
2. We propose an adaptive, perception-based method for degeneracy analysis and mitigation using high-localizability features from the BR. The degeneracy indices for LiDAR SLAM are mathematically defined, and the degeneracy state is analyzed in real-time based on the information matrix from environmental observations. A high-localizability residual is then constructed from the BR, enabling effective degeneracy mitigation and robust state estimation.
3. We verify the feasibility and superiority of the proposed method in multiple real-world degraded tunnels. Experimental results show that the integration of high-localizability feature constraints from multiple robots can compensate for degeneracy in various typical degraded tunnels, demonstrating significant advantages in robust and accurate trajectory localization and high-fidelity global and local mapping over state-of-the-art representative algorithms. This fully verifies the applicability of the proposed framework in real degraded tunnel environments.

Section 2 reviews relevant SLAM and multi-robot systems. Section 3 details our three-module framework integrating degeneracy analysis, high-localizability feature from leapfrogging collaboration, and degeneracy mitigation. Section 4 validates performance across three tunnels with comparative analysis. Section 5 summarizes contributions, discusses limitations, and proposes extensions.

2. Related work

The development of SLAM technology covers multiple directions, among which the research closely related to this work mainly includes three aspects: (i) LiDAR-inertial odometry for ego-motion estimation, (ii) degeneracy-aware methods to detect and mitigate geometric failure, and (iii) multi-robot collaboration to compensate for observability limits.

2.1. LiDAR-inertial odometry

LiDAR-inertial odometry provides high-precision ego-motion estimation via tightly-coupled optimization, but its performance collapses in degraded tunnel environments where sparse geometric structure yields insufficient correspondences [21,22]. Early point-registration methods suffer from initialization sensitivity and quadratic computational complexity [23,24], while feature-based approaches like LOAM [6] and LeGO-LOAM [7] extract planes and edges. In repeating tunnel geometries, however, these handcrafted features become indistinguishable, causing the information matrix to approach singularity and rendering odometry drift inevitable [11].

Even with IMU integration, which provides short-term drift resistance, LiDAR-inertial methods remain vulnerable to geometric degeneracy in tunnels. LIO-SAM [8] employs factor-graph optimization coupling IMU pre-integration with loop closures, but its global consistency hinges on detectable loops—unavailable in long, straight tunnels where drift accumulates unchecked. By leveraging terrestrial constraints and sliding window consistency, Duan et al. [25] introduced a point-wise dynamic removal method within the LIO framework to eliminate ghost trails and reduce pose estimation drift. Fast-LIO [26] and Fast-LIO2 [9] adopt ESIKF [27] and point-plane matching for computational efficiency, but repetitive planar surfaces in tunnels induce matching ambiguities that corrupt the Kalman innovation sequence. KISS-ICP [28] employs a simple yet robust point-to-point ICP framework with a constant velocity motion model and an adaptive thresholding strategy, achieving state-of-the-art odometry performance without the need for complex feature extraction or deskewing. Point-LIO [29] is a highly efficient LiDAR-inertial odometry that directly operates on raw point clouds through a per-point EKF update, effectively bypassing the computational overhead of local map construction and feature tracking. While D-LIO [30] and OR-LIM [31] address dynamic motion artifacts, they do not remedy the fundamental rank deficiency of the observability matrix caused by sparse cross-sectional geometry. Consequently, multi-sensor fusion merely postpones, rather than prevents, inevitable odometric failure in degraded tunnel environments—a conclusion supported by our baseline experiments showing Fast-LIO2 exceeding 17 m of start-to-end error over three experimental scenes (see Fig. 11).

2.2. Degeneracy-aware SLAM

Degeneracy-aware SLAM methods detect and mitigate rank deficiency of the information matrix in degraded tunnel environments. Tuna et al. [14] and Zhang et al. [32] use singular value decomposition to identify ill-posed optimization and apply regularization to suppress distortion of eigenvalue distribution. However, these approaches are fundamentally reactive: they can only attenuate existing information loss but cannot synthesize new geometric constraints where LiDAR returns are spatially aliased. Consequently, in feature-devoid tunnels where the observability matrix is intrinsically rank-deficient, degeneracy-aware methods merely delay divergence rather than enable robust long-term navigation.

Zhang et al. [12] established the theoretical foundation for degeneracy detection via information matrix rank deficiency. Subsequent approaches attempt within single-robot frameworks remain fundamentally limited: [13] reactively downweight unreliable constraints without synthesizing new information; X-ICP [14] employs regularization to impose artificial stiffness, biasing state estimates; Hatleskog and Alexis [33] relies on point-plane matching that exacerbates spatial aliasing in repetitive tunnel geometries.

Confined to internal constraint manipulation, these methods merely attenuate rather than resolve observability deficits in degraded tunnels, making external geometric constraint injection indispensable.

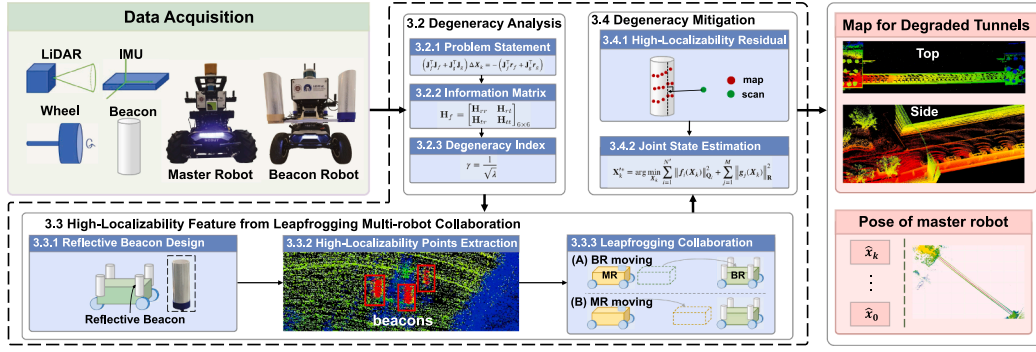


Fig. 1. System architecture of Leap-SLAM.

2.3. Multi-robot SLAM

Multi-robot SLAM enhances mapping efficiency and global consistency in large-scale environments via collaborative optimization [16,34]. However, in GNSS-denied tunnels, existing collaboration strategies—predicated on detectable loop closures—merely shift the bottleneck from observability to detection reliability, rather than resolving tunnel-specific degeneracy.

Recent efforts optimize multi-robot collaboration via distributed architectures and lightweight information sharing [17,18,35,36], but remain fundamentally anchored to detectable inter-robot loop closures. Swarm-SLAM [17] and DiSCo-SLAM [18] rely on local feature descriptors and Scan Context for loop detection, which become indistinguishable in repetitive tunnel geometries, causing false constraints or detection failure. LAMP [11,37] assume detectable inter-robot loops for graph optimization—precisely what’s absent in feature-devoid tunnels. Zhang et al. [19] integrates real-time anchor detection to improve consistency, but anchor recall drops significantly in feature-devoid tunnels, rendering the method unreliable. Lightweight architectures [35] and benchmark datasets [36] address computational efficiency and evaluation logistics, but cannot compensate for the fundamental information deficit. Consequently, these methods optimize collaboration logistics without remedying core observability limits, making them ill-suited for robust long-term navigation in degraded tunnel environments.

Despite advances in single-robot odometry, degeneracy-aware, degeneracy-mitigated, and multi-robot collaboration, existing methods remain fundamentally constrained by their reliance on detectable environmental features or computable inter-robot loops—both of which fail in degraded tunnels. This work proposes Leap-SLAM, a degeneracy-mitigated robust framework that breaks this reliance by actively generating high-localizability point constraints through leapfrogging collaboration between a beacon robot and a master robot, thereby ensuring stable, long-term localization where existing methods inevitably diverge.

3. Approach

The proposed Leap-SLAM framework is implemented based on the Fast-LIO2 [9] system, inheriting its robust tightly-coupled LiDAR-inertial odometry and efficient ikd-Tree-based map management. We extend this foundational architecture by introducing a leapfrogging collaborative strategy and an information-matrix-driven degeneracy mitigation mechanism, specifically tailored for extreme feature-starved tunnel environments.

3.1. Overview

This section presents the systematic architecture of Leap-SLAM, providing an overview of the proposed framework and its key components.

The proposed framework, Leap-SLAM, is specifically designed for degraded tunnels which lack sufficient geometric constraints, employing a

pose estimation method based on ESIKF and a degeneracy analysis approach derived from information matrix decomposition. Departing from classical symmetric multi-robot SLAM, our system introduces a multi-robot SLAM paradigm that achieves both collaborative localization and consistent 3D mapping for the team. The system consists of a MR responsible for localization and mapping, and a BR that provides supplementary geometric constraints. This architecture eliminates the need for redundant information exchange or map merging, thereby maximizing system efficiency and robustness in degraded tunnels. The framework is structured around three core modules.

Fig. 1 illustrates three modules along with the estimated pose of the MR and the corresponding point cloud in degraded tunnel environments. (i) The degeneracy analysis module, under the problem statement in the paper, constructs an information matrix from LiDAR observations and derives the degeneracy index to quantify the degree of scene degeneracy, providing real-time evaluation of degeneracy status as the basis for external features supplement and degeneracy mitigation. (ii) The high-localizability feature extraction module mounts reflective beacons on the BR to facilitate the extraction of high-localizability feature points, which serve as reliable external features based on leapfrogging multi-robot collaboration. (iii) The degeneracy mitigation module focuses on the MR’s pose estimation, integrating high-localizability constraints into the ESIKF framework to achieve accurate and robust pose estimation under degenerate conditions. The sequential poses estimated for the MR are then used to integrate successive LiDAR scans, resulting in a high-definition, geometric-precise global point cloud map of degraded tunnels.

3.2. Degeneracy analysis

The core challenge in degraded tunnels lies in repetitive and weak geometric structures, which lead to insufficient reliable constraints. The degeneracy analysis module builds an information matrix based on LiDAR constraints and applies eigenvalue decomposition to obtain degeneracy indices, which characterize the extent of environmental degeneracy. The obtained results serve as the basis for determining whether to utilize the high-localizability residual (details in Section 3.4.1).

3.2.1. Problem statement

This section begins by formulating the SLAM problem for degraded tunnels to establish a basis for the subsequent degeneracy analysis. We adaptively estimate robot pose and map using tightly-coupled LiDAR-inertial measurements, augmented by actively generated high-localizability beacon constraints from leapfrogging collaboration to compensate for insufficient natural geometric features.

Referring to the framework of Fast-LIO2 [9], the proposed framework adopts ESIKF with the tightly-coupled LiDAR-inertial odometry for state estimation. Specifically, the SLAM problem is described as follows: let the state of the robot at time k be $\mathbf{X}_k$:

$$\mathbf{X}_k = [\mathbf{R}_k^T, \mathbf{p}_k^T]^T, \quad (1)$$

where $\mathbf{R}_k^T$ and $\mathbf{p}_k^T$ represent the orientation and position of the LiDAR at time k , respectively. Then, we aim to find an optimal state $\mathbf{X}_k^*$:

$$\begin{aligned} \mathbf{X}_k^* &= \arg \min_{\mathbf{X}_k} F, \\ f_i(\mathbf{X}_k) &= d_i, \\ F &= \sum_{i=1}^N \|f_i(\mathbf{X}_k)\|^2. \end{aligned} \quad (2)$$

The optimal state $\mathbf{X}_k^*$ is estimated by jointly minimizing the point-plane residuals F and the IMU motion residuals G , which are defined similarly. Given that this study primarily addresses the structural degeneracy of LiDAR SLAM, this section focuses on the LiDAR-related components, with detailed descriptions of IMU residuals and constraints abbreviated for brevity.

Referring to the degeneracy-aware method for SLAM problems proposed by [38], this paper adopts the Gauss-Newton method [39] to solve the incremental nonlinear least squares problem of the LiDAR constraint F through iteration:

$$\underbrace{\sum_{i=1}^N \mathbf{J}_i^T(\mathbf{X}_k) f_i(\mathbf{X}_k)}_{-\mathbf{b}} + \underbrace{\sum_{i=1}^N \mathbf{J}_i^T(\mathbf{X}_k) \mathbf{J}_i(\mathbf{X}_k) \Delta \mathbf{X}_k}_{\mathbf{H}_f(\mathbf{X}_k)} = 0, \quad (3)$$

where $\Delta \mathbf{X}_k$ denotes the state increment in the k -th iteration, which is used to update the current state $\mathbf{X}_k$. $\mathbf{J}_i(\mathbf{X}_k)$ is the Jacobian matrix of $f_i(\mathbf{X}_k)$. The Jacobian matrix of the residual is:

$$\mathbf{J}_i(\mathbf{X}_k) = \frac{\partial f_i(\mathbf{X}_k)}{\partial \mathbf{X}_k} = \left[\frac{\partial f_i}{\partial \mathbf{R}}, \frac{\partial f_i}{\partial \mathbf{t}} \right]. \quad (4)$$

Based on Gauss-Newton iteration, the incremental optimization form of the least squares problem can be uniformly expressed as:

$$\left(\mathbf{J}_f^T \mathbf{J}_f + \mathbf{J}_g^T \mathbf{J}_g \right) \Delta \mathbf{X}_k = - \left(\mathbf{J}_f^T \mathbf{r}_f + \mathbf{J}_g^T \mathbf{r}_g \right) \triangleq \mathbf{H} \Delta \mathbf{X}_k = \mathbf{b}, \quad (5)$$

Thus, the total information matrix $\mathbf{H}$ and vector $\mathbf{b}$ integrate the geometric constraints from LiDAR, forming a unified optimization framework.

3.2.2. Information matrix

We construct the information matrix from LiDAR residuals, while IMU terms follow the same derivation but are omitted for brevity. Under the second-order approximation, the LiDAR information matrix $\mathbf{H}_f$ is expressed as:

$$\begin{aligned} \mathbf{H}_f &= \sum_{i=1}^n \mathbf{J}_i^T(\mathbf{X}_k) \mathbf{J}_i(\mathbf{X}_k) \\ &= \begin{bmatrix} \mathbf{H}_{rr} & \mathbf{H}_{rt} \\ \mathbf{H}_{tr} & \mathbf{H}_{tt} \end{bmatrix}_{6 \times 6}, \end{aligned} \quad (6)$$

where $\mathbf{H}_{rr}$ and $\mathbf{H}_{tt}$ respectively reflect the observation constraint strengths in the rotational and translational directions.

Algorithm 1 formalizes the information matrix construction process using normal points $\mathcal{X}_n$, high-localizability points $\mathcal{X}_h$ (details in Eq. (10)), and the current robot states as inputs. It outputs a normal information matrix and a high-localizability-constrained information matrix.

3.2.3. Degeneracy index

The information matrix of rotation and translation characterizes the predicted state of the SLAM system through eigenvalue decomposition on $\mathbf{H}_{rr}$ and $\mathbf{H}_{tt}$ respectively:

$$\mathbf{H}_{rr} = \mathbf{V}_r \boldsymbol{\Sigma}_r \mathbf{V}_r^T = \mathbf{V}_r \begin{bmatrix} \lambda_{r1} & 0 & 0 \\ 0 & \lambda_{r2} & 0 \\ 0 & 0 & \lambda_{r3} \end{bmatrix} \mathbf{V}_r^T, \quad (7)$$

$$\mathbf{H}_{tt} = \mathbf{V}_t \boldsymbol{\Sigma}_t \mathbf{V}_t^T = \mathbf{V}_t \begin{bmatrix} \lambda_{t1} & 0 & 0 \\ 0 & \lambda_{t2} & 0 \\ 0 & 0 & \lambda_{t3} \end{bmatrix} \mathbf{V}_t^T, \quad (8)$$

Algorithm 1 Information matrix construction.

Input: normal points $\mathcal{X}_n$, high-localizability points $\mathcal{X}_h$, current state $\mathbf{X}$
Output: information matrix $\mathbf{H}_f$, information matrix with high-localizability feature constraints $\mathbf{H}'_f$

```

1:  $\mathbf{H}_f \leftarrow [0, 0, 0, 0, 0, 0]$ 
2: for all  $p$  in  $\mathcal{X}_n$  do
3:    $\mathbf{J} \leftarrow f(p, \mathbf{X})$ 
4:    $\mathbf{H}_f \leftarrow \mathbf{H}_f + \mathbf{J}^T \mathbf{J}$ 
5: end for
6: if isDegenerate then
7:    $\mathbf{H}'_f \leftarrow \mathbf{H}_f$ 
8:   for all  $p$  in  $\mathcal{X}_h$  do
9:      $\mathbf{J} \leftarrow f(p, \mathbf{X})$ 
10:     $\mathbf{H}'_f \leftarrow \mathbf{H}'_f + \mathbf{J}^T \mathbf{J}$ 
11:   end for
12: else
13:   return  $\mathbf{H}_f$ 
14: end if
15: return  $\mathbf{H}_f, \mathbf{H}'_f$ 
```

where $\lambda_{t1}, \lambda_{t2}, \lambda_{t3}$ denote the eigenvalues corresponding to the translational degeneracy indices along dominant directions, and eigenvalues are sorted as $\lambda_{r1} > \lambda_{r2} > \lambda_{r3}$ and $\lambda_{t1} > \lambda_{t2} > \lambda_{t3}$. Each eigenvector defines an independent observation direction, with its eigenvalue quantifying constraint strength [14]; smaller eigenvalues indicate weaker constraints along that direction. Typically, the smallest eigenvalue λ_{t3} corresponds to the direction parallel to the tunnel axis at the robot's current position, reflecting weak geometric constraints along this dimension. Each computation is utilized for real-time assessment to determine whether degeneracy is occurring at the current instant. In the practical applications, due to the differences in scale and type between rotation and translation, the scale differences of eigenvalue λ_i in different dimensions are quite significant. Therefore, we further normalize them and define the degeneracy indices for rotation and translation respectively:

$$\gamma = \frac{1}{\sqrt{\lambda}}, \quad (9)$$

where γ represents the uncertainty of pose estimation along the corresponding eigenvector direction. Larger γ values directly correlate with increased estimation errors and pronounced degeneracy in that dimension. Based on experimental validation in tunnel environments, a degeneracy threshold $\tau = 0.15$ is established; when $\gamma > \tau$, the system activates degeneracy-aware state estimation strategies (Section 4.2.1).

In summary, the degeneracy analysis module provides real-time outputs of the scene's degeneracy extent and direction vector, offering valuable environmental prior information for localization degeneracy mitigation and multi-robot collaboration.

3.3. High-localizability feature from leapfrogging multi-robot collaboration

The high-localizability feature points provided by the BR serve as supplementary external constraints, enabling Leap-SLAM to achieve accurate and robust localization in geometrically degraded tunnels. Despite the significant advancements made by state-of-the-art SLAM frameworks [8,9], these methods remain susceptible to performance degeneracy in environments with repetitive or sparse geometric structures due to the paucity of reliable constraints.

To address this fundamental limitation, this paper introduces high-localizability features to provide external observations for the pose estimation of the MR, specifically targeting the compensation for constraint deficits along the longitudinal tunnel direction. Furthermore, a suite of degeneracy indices for LiDAR SLAM is formulated based on the information matrix derived from environmental observations, providing a quantitative measure for real-time degeneracy assessment.

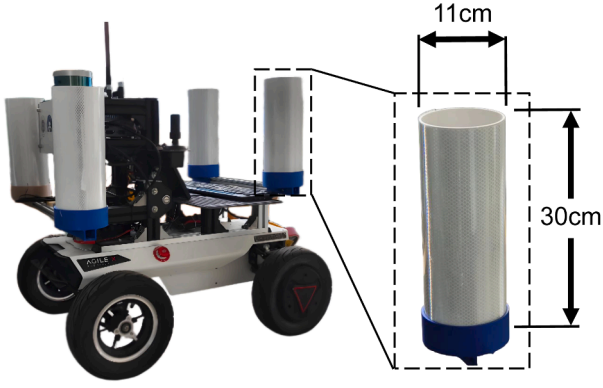


Fig. 2. Beacon robot with four reflective cylindrical beacons (diameter: 11 cm, height: 30 cm) mounted on its periphery.

3.3.1. Reflective beacon design

The reflective beacons carried by the BR aim to provide an easily identifiable, stable and reliable observation. In degraded tunnels, the environmental structure can provide fundamental constraints for rotation, lateral translation, and vertical translation, but the translation along the tunnel axis often lacks an effective reference, becoming the main source of error accumulation. Therefore, the beacon design must meet two requirements simultaneously: firstly, to ensure high distinguishability in complex point clouds; secondly, to provide strong, stable and reliable constraints, especially in the tunnel translational direction.

The proposed framework employs four cylindrical beacons (11 cm in diameter and 30 cm in height for each) mounted symmetrically on BR. As illustrated in Fig. 2, each beacons are covered with a high-reflectivity material, and four identical beacons are set around the robot. LiDAR observations of the cylindrical beacon reveal two salient characteristics: first, its distinct cylindrical geometry induces a structured point cloud distribution; second, its high reflectivity yields intensity returns that exhibit pronounced contrast against the background, enabling robust feature extraction. Consequently, the synergistic integration of these geometric and radiometric properties equips the beacon to deliver persistent and discriminative geometric constraints within tunnel environments.

3.3.2. High-localizability points extraction

The goal of feature point extraction is to identify high-localizability constraints from the original point cloud $\mathbf{P}$ via a multi-stage pipeline consisting of intensity-based candidate selection, spatial clustering, and geometric prior-model verification. This explicit classification into a high-localizability set $\mathcal{X}_h$ and a normal environmental set $\mathcal{X}_n$ is essential for three primary reasons. Firstly, it enables the ESIKF to assign asymmetric noise covariances, ensuring high-precision geometric anchors are not overwhelmed by degraded environmental features. Secondly, it prevents sparse beacon points from being erroneously discarded during voxel downsampling. Thirdly, it facilitates the automatic removal of the BR's footprint, ensuring a clean global map without complex post-processing. Consequently, this strategy maintains both the sparsity of informative constraints and the global consistency of the mapping results.

Firstly, the high-localizability feature point set $\mathcal{X}_h$ and the normal environmental points $\mathcal{X}_n$ are extracted from the original point cloud $\mathbf{P}$ through a sequential process of intensity filtering and geometric verification:

$$\begin{cases} \mathcal{X}_c = \{\mathbf{p}_i \in \mathbf{P} \mid I(\mathbf{p}_i) \geq \xi\} \\ \mathcal{X}_h = \{C_j \mid |H_j - H_{ref}| < \epsilon_H, L_j > \tau_L\}, \\ \mathcal{X}_n = \mathbf{P} \setminus \mathcal{X}_h = \{\mathbf{p}_i \in \mathbf{P} \mid \mathbf{p}_i \notin \mathcal{X}_h\} \end{cases} \quad (10)$$

where $\mathcal{X}_c$ is the candidate set filtered by reflection intensity $I(\mathbf{p}_i)$ with threshold ξ . To reject interference from environmental reflectors

like road signs, an Euclidean clustering algorithm is applied to $\mathcal{X}_c$, partitioning it into discrete clusters $\{C_1, C_2, \dots, C_k\}$. Each cluster C_j is validated against the physical dimensions of the portable beacons; specifically, its axial height H_j and cylindrical linearity L_j (derived via PCA) must satisfy the constraints relative to the reference height H_{ref} and threshold τ_L . Finally, the environmental point set $\mathcal{X}_n$ is defined as the complement of the identified high-localizability features.

By introducing high-localizability feature points, the BR compensates for the lack of natural geometric constraints. Fig. 3 presents LiDAR point clouds of the BR operating in a forested environment exterior to the tunnel, observed by the MR. The visualization sequence progresses from a bird view (left) to local views centered on the BR. High-intensity points on the BR exhibit pronounced reflectance contrast against the background, enabling reliable identification and segmentation. These features correspond precisely to the reflective beacons mounted on the BR's periphery. The high-localizability points extraction module further ensures strong, stable and reliable observation constraints, thereby providing robust support for localization in degraded tunnel environments.

3.3.3. Leapfrogging collaboration

After proposing the extraction method for high-localizability points, we introduce a leapfrogging collaborative advancement strategy. Following the Fast-LIO2 [9] methodology, the system fuses wheel odometry with the posterior velocity estimate $\mathbf{v}_l$ derived from IMU pre-integration and LiDAR-inertial optimization. Specifically, $\mathbf{v}_l$ is estimated by integrating IMU measurements for forward propagation and iteratively refining the state through LiDAR-based manifold constraints. To mitigate potential wheel slippage, a dual-threshold stationary state detection is implemented. We define a stationary state indicator $\mathcal{I}$ based on the joint consistency between the wheel odometry velocity v_W and the magnitude of the filtered velocity $\|\mathbf{v}_l\|$:

$$\mathcal{I} = \begin{cases} 1, & \text{if } \max(v_W, \|\mathbf{v}_l\|) < \delta \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

where δ denotes a predefined threshold for zero-velocity consistency. This strategy ensures that the MR can consistently observe the beacons on the BR and obtain stable high-localizability points. By the coordination of the MR and BR, and taking the basic motion unit as the movement distance unit, the beacons on the BR are always kept within the observation range of the MR. This information fusion approach prevents potential odometry misjudgments and provides a reliable, drift-resistant data foundation for subsequent joint state estimation.

The leapfrogging collaboration strategy divides motion into basic motion units (specific length as detailed in Section 4.2.2), forming a closed-loop cooperative pattern through the alternating movement of the BR and MR. As illustrated in Fig. 4, both robots first remain stationary for initialization. Then, the BR moves forward by one motion unit along the preset path and enters the residence mode, acting as a new beacon to provide localization constraints for the MR. Subsequently, the MR advances by the same distance to restore the initial spacing, completing one collaborative cycle. In this mode, the two robots advance in a manner of two frogs alternately leaping across a pond: one robot remains stationary as a stable anchor while the other moves forward. This alternating process repeats until the entire tunnel is traversed. Once the degeneracy analysis module detects recovery from the degenerate state, the leapfrogging collaboration strategy is automatically deactivated. Notably, to ensure stable relative observations and maintain the BR within the MR's FOV, the movements of the robots are currently managed through manual remote control. For safety, the BR is equipped with a TEB local planner [40] for obstacle avoidance, which is configured for emergency stops based on localized sensor data and operates independently of the MR's SLAM map. This implementation allows the study to focus on the core collaborative SLAM framework and degeneracy mitigation.

In addition, to avoid incorrectly matching dynamic objects such as pedestrians as static structures, which would generate erroneous rela-

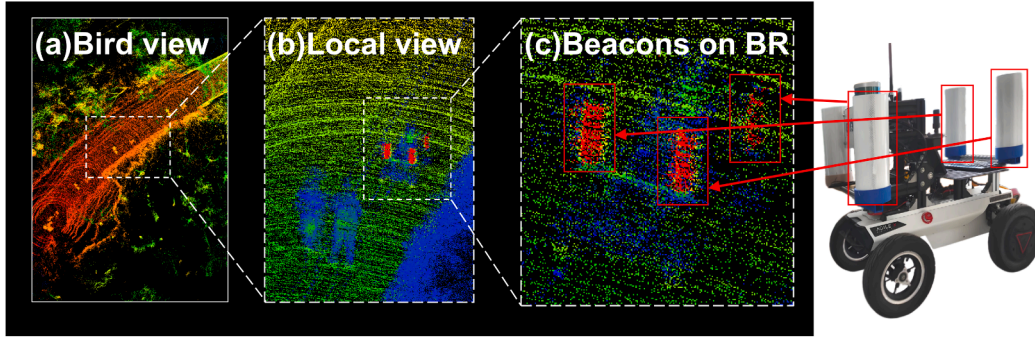


Fig. 3. LiDAR point cloud for reflective beacons. (a) bird view. (b) local view of Beacon Robot. (c) magnified view for reflective beacons on BR. Points (red points) with high intensity correspond to high-localizability features.

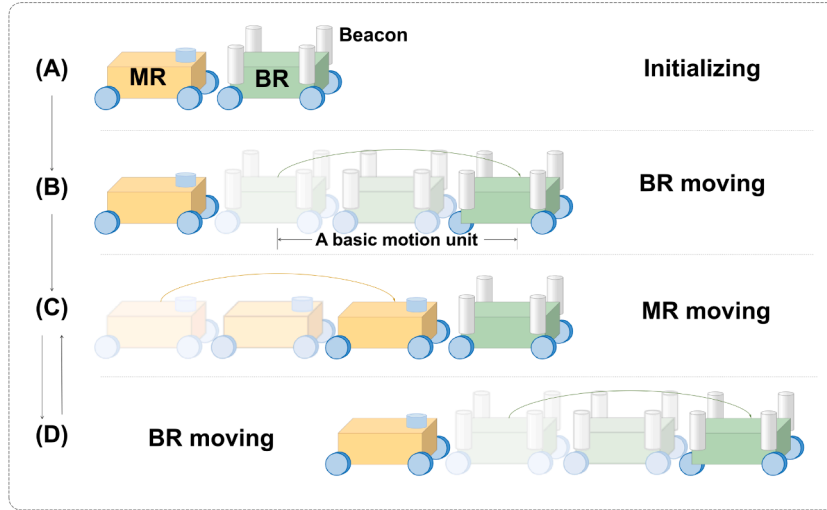


Fig. 4. Illustration of one cycle of the leapfrogging cooperative strategy. The BR and MR alternately perform advance-and-stay operations. (A) BR departs from the starting point and moves one basic motion unit. (B) MR follows the BR's path for one basic motion unit. (C) Back to (A), BR departs again. (D) Back to (B).

tive pose estimation, this module designs a SLAM freezing mechanism based on the robot's motion state. The current motion state of the MR is determined by cross-validating the chassis odometry with IMU acceleration and angular velocity data. This approach ensures that the stationary state is accurately identified even in cases of potential wheel slippage. If the robot is confirmed to be in a stationary state, the pose update process of the MR is immediately frozen, the SLAM process is paused, and the pose output of the previous frame is reused. This strategy effectively eliminates LiDAR matching interference from dynamic objects during periods of physical immobility, further enhancing the system's reliability in field environments.

3.4. Degeneracy mitigation

Under the ESIKF framework [27], high-localizability feature points are incorporated as external observations to enhance pose estimation in directions with weak geometric constraints. The core of this process lies in the formulation of observation models and noise representations, where both kinematic and measurement uncertainties are explicitly modeled. Higher reliability weights are assigned to the high-localizability feature constraints, thereby facilitating effective degeneracy mitigation.

3.4.1. High-localizability residual

In Section 3.2.1, the LiDAR and IMU constraints are theoretically formulated. However, in practical scenarios, observation and system noises inevitably introduce deviations into these constraints. The following section focuses on the error modeling under LiDAR observation constraints.

Based on the classification described in Section 3.3.2, the LiDAR residual function is formulated as:

$$f_i(X_k) = \begin{cases} d_i, & \mathbf{p}_i \in \mathcal{X}_n \\ d_{hi}, & \mathbf{p}_i \in \mathcal{X}_h \end{cases}, \quad (12)$$

where $\mathcal{X}_n$ and $\mathcal{X}_h$ denote the sets of normal environmental points and high-localizability points, respectively. The residual error is then defined as:

$$u_i = f_i(X_k^*) - \begin{cases} d_i, & \mathbf{p}_i \in \mathcal{X}_n \\ d_{hi}, & \mathbf{p}_i \in \mathcal{X}_h \end{cases} \sim \mathcal{N}(0, \mathbf{Q}_i), \quad (13)$$

where $\mathbf{Q}_i$ represents the observation noise covariance matrix. During real-time operation, our system dynamically monitors degeneracy using the index γ (Section 3.2.3). Upon detecting degeneracy ($\gamma > \tau$), high-localizability points are incorporated into the state estimation.

3.4.2. Joint state estimation

When the degeneracy analysis module identifies a degenerate condition, high-localizability feature constraints are incorporated into the system. The residual function of these constraints is defined as:

$$F' = \sum_{i=1}^{N'} \|f_i(X_k)\|_{\mathbf{Q}_i}^2, \quad (14)$$

and the corresponding information matrix is given by:

$$\mathbf{H}'_f = \sum_{i=1}^{N'} \mathbf{J}_i^T(X_k) \mathbf{J}_i(X_k), \quad (15)$$

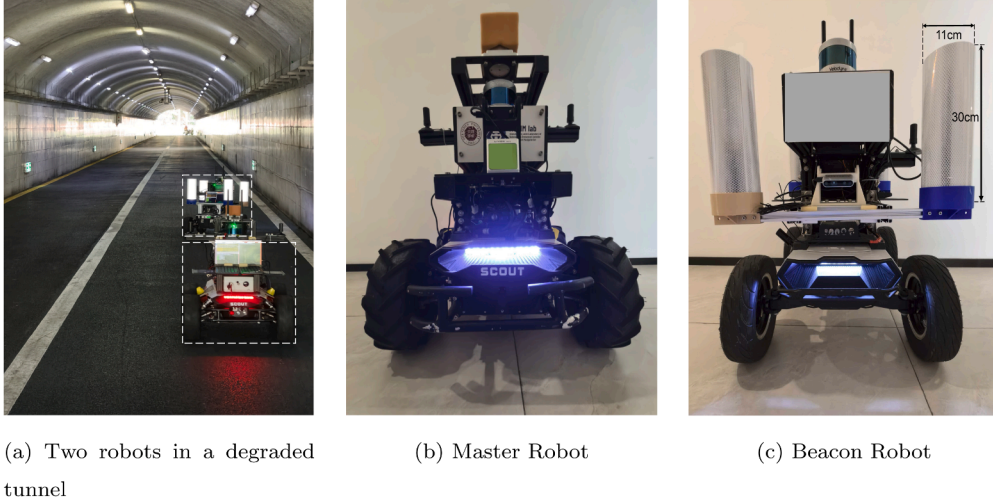


Fig. 5. Experimental robotic platform.

where $\mathbf{J}'_i$ denotes the Jacobian matrix that integrates the high-localizability point constraints, formulated as:

$$\mathbf{J}'_i(\mathbf{X}_k) = \begin{bmatrix} \mathbf{J}_i(\mathbf{X}_k) \\ \mathbf{J}_{h,i}(\mathbf{X}_k) \end{bmatrix}. \quad (16)$$

The final state estimation is formulated as a joint optimization problem under the ESIKF framework:

$$\mathbf{X}_k^{fs} = \arg \min_{\mathbf{X}_k} \sum_{i=1}^{N'} \|\mathbf{f}_i(\mathbf{X}_k)\|_{\mathbf{Q}_i}^2 + \sum_{j=1}^M \|\mathbf{g}_j(\mathbf{X}_k)\|_{\mathbf{R}}^2, \quad (17)$$

where the first term represents LiDAR constraints augmented with high-localizability features, and the second term corresponds to the IMU motion constraints. By employing iterative Kalman filtering, the system efficiently converges to the optimal state estimate $\mathbf{X}_k^{fs}$. To fully leverage the high-localizability capability of the reflective beacons, the observation noise covariance for these points is preset to a value of 0.1 times that of normal points. This configuration is grounded in the fact that the LiDAR's measurement error remains constant and stable within its effective operating range (up to 100 m), justifying a non-dynamic, fixed covariance setting. Furthermore, since these reflective features provide significantly more reliable and deterministic spatial information than cluttered environmental points, assigning a smaller noise level allows the system to enforce tighter constraints, effectively mitigating localization drift and ensuring robustness in degraded tunnel environments.

4. Experiment

4.1. Experimental setup

The experimental platform consists of two Agilex mobile robots—a Scout mobile robot (master robot) and a Hunter mobile robot (beacon robot)—as illustrated in Fig. 5. The master robot is equipped with a Velodyne VLP-16 LiDAR, chassis wheel encoders, and an SBG Ellipse-N IMU. The Hunter robot is equipped with a single-antenna GNSS receiver, which provides centimeter-level RTK positioning, serving as the ground-truth reference for tunnel traversal endpoints. Onboard computing resources include an Intel i7-1260P CPU and 16 GB of RAM. Four reflective beacons (Fig. 5(c)) are mounted on the Hunter robot to enable easily identifiable, stable and reliable external observations.

Three long tunnels are selected as the experimental datasets: a 200-meter traffic tunnel, a 1000-meter traffic tunnel, and a 200-meter pedestrian tunnel, labeled Tunnel 1, Tunnel 2, and Tunnel 3, respectively, as illustrated in Fig. 6. The experimental scenes feature sufficiently rough

Table 1

Key parameter configurations of leap-SLAM.

Parameter	Value	Description
$\mathbf{p}_{IL}$	[0, 0, 0.28]	Translation from LiDAR to IMU (m)
$\mathbf{R}_{IL}$	$\mathbf{I}_{3 \times 3}$	Rotation from LiDAR to IMU
ξ	250	Intensity threshold for reflective beacon candidates
τ	0.15	Threshold for geometric degeneracy detection
$\mathbf{Q}_h$	$0.1\mathbf{Q}_i$	Covariance for high-localizability features
D_{beacon}	0.11 m	Physical diameter of the reflective beacon
H_{beacon}	0.30 m	Physical height of the reflective beacon
d_{min}	0.2 m	Blind zone radius for LiDAR point filtering
l_{voxel}	0.2 m	Voxel size for surface feature downsampling

ground surfaces that provide reliable friction for the mobile platforms. Benefiting from these favorable surface conditions and smooth motion control, no observable wheel slippage occurred during the data collection process, thereby ensuring the integrity of the wheel odometry for stationary state detection. Each dataset contains LiDAR, IMU, wheel odometry, and RTK data from the MR, with RTK measurements used exclusively for start-to-end trajectory evaluation. Data collection follows the leapfrogging collaboration strategy, starting at one tunnel entrance, proceeding through the tunnel, and exiting at the opposite end. The experimental evaluation mainly consists of three parts in this study. Firstly, we investigate the feasibility of performing degeneracy analysis and identifying high-localizability feature points within degraded tunnel environments. Secondly, we further evaluate the accuracy of the proposed framework under degraded tunnel conditions. Thirdly, a rational analysis of the degeneracy-mitigated mechanism is conducted.

To ensure the reproducibility of the experimental results and provide a comprehensive overview of the system configuration, the key parameters of Leap-SLAM are summarized in Table 1.

Specifically, $\mathbf{p}_{IL}$ and $\mathbf{R}_{IL}$ denote the calibrated translation and rotation extrinsics between the LiDAR and IMU sensors. The intensity threshold ξ is set to 250 for filtering reflective beacon candidates, while the geometric degeneracy detection threshold τ and the high-localizability feature covariance $\mathbf{Q}_h$ are determined based on the theoretical framework in Section 3. The physical dimensions of the beacon, including its diameter D_{beacon} and height H_{beacon} , are explicitly defined for geometric model matching. Furthermore, d_{min} and l_{voxel} , representing the LiDAR blind zone radius and the voxel size for surface feature downsampling, are essential configurations inherited from the Fast-LIO2 framework. These parameters are critical as they directly influence the underlying odometry performance and the resulting point cloud density.



Fig. 6. Experimental scenes employed for evaluation. The three scenes, arranged left to right, are a 200-meter road tunnel, a 1000-meter road tunnel, and a 200-meter pedestrian tunnel.

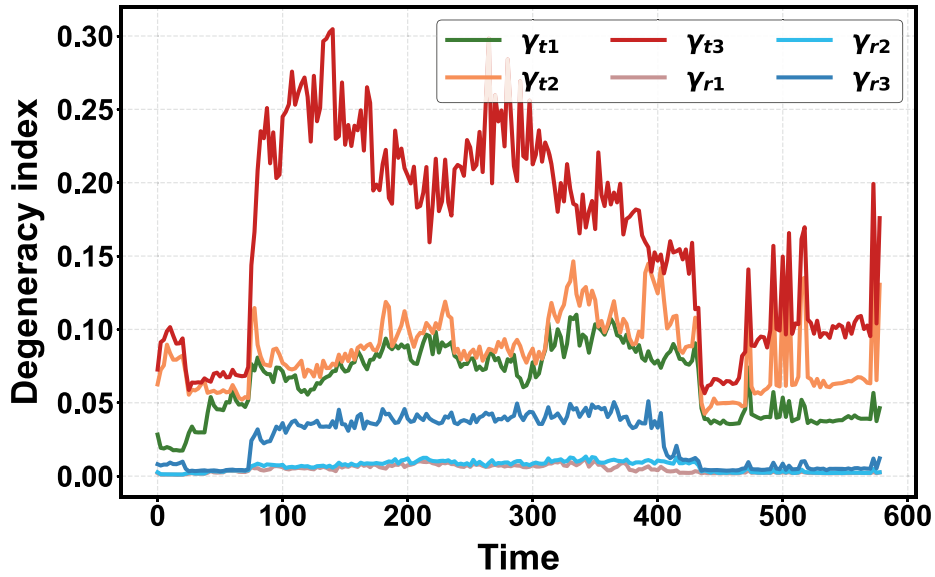


Fig. 7. Temporal distribution of six-DoF degeneracy indices in Tunnel 3. Robot enters tunnel at 90 s, exits at 400 s.

4.2. Degeneracy analysis and feature extraction

4.2.1. Degeneracy analysis in tunnels

This experiment investigates the feasibility of degeneracy analysis proposed in Section 3.2.3 and evaluates the degeneracy state within tunnel environments. Tunnel 3 is selected as the representative case, where the master robot's normal points are utilized to compute the corresponding information matrix and degeneracy index following Algorithm 1 and Eq. (9). The temporal distribution of six-DoF degeneracy indices in Tunnel 3 is presented in Fig. 7. The value γ_{t3} in a certain degree of freedom $t3$ exhibits significantly higher values compared to the other five degrees of freedom, with its direction aligned along the tunnel's forward translation axis. Notably, γ_{t3} surpasses 0.15 throughout 90–400 s, corresponding to tunnel entry/exit phases. Spatial distribution of γ_{t3} employs trajectory-based color mapping and point cloud rendering depicted in Fig. 8. These results demonstrate that MR's SLAM using environmental normal points suffers severe degeneracy in forward translation within tunnels, mirroring observations in the other two tunnels. The analysis

establishes a degeneracy threshold $\tau = 0.15$, classifying states as degenerate when indices exceed this value.

Overall, repetitive tunnel structures and insufficient feature constraints result in persistently high degeneracy indices, especially along the tunnel's forward translation axis, thereby highlighting the feasibility for degeneracy analysis and requiring the incorporation of high-localizability feature points as compensatory constraints.

4.2.2. High-localizability feature extraction

This study evaluates the identification performance of high-localizability feature points as a function of distance. Designating the inter-device distance between the Master Robot (MR) and Beacon Robot (BR) as the primary experimental variable, we systematically quantified LiDAR point cloud features exhibiting a reflection intensity $I \geq \xi$. This threshold ξ , empirically set to 250 in our implementation, provides a quantitative evaluation of high-localizability point extraction efficacy by effectively isolating highly reflective beacon features from the background environment. The experimental validation was conducted

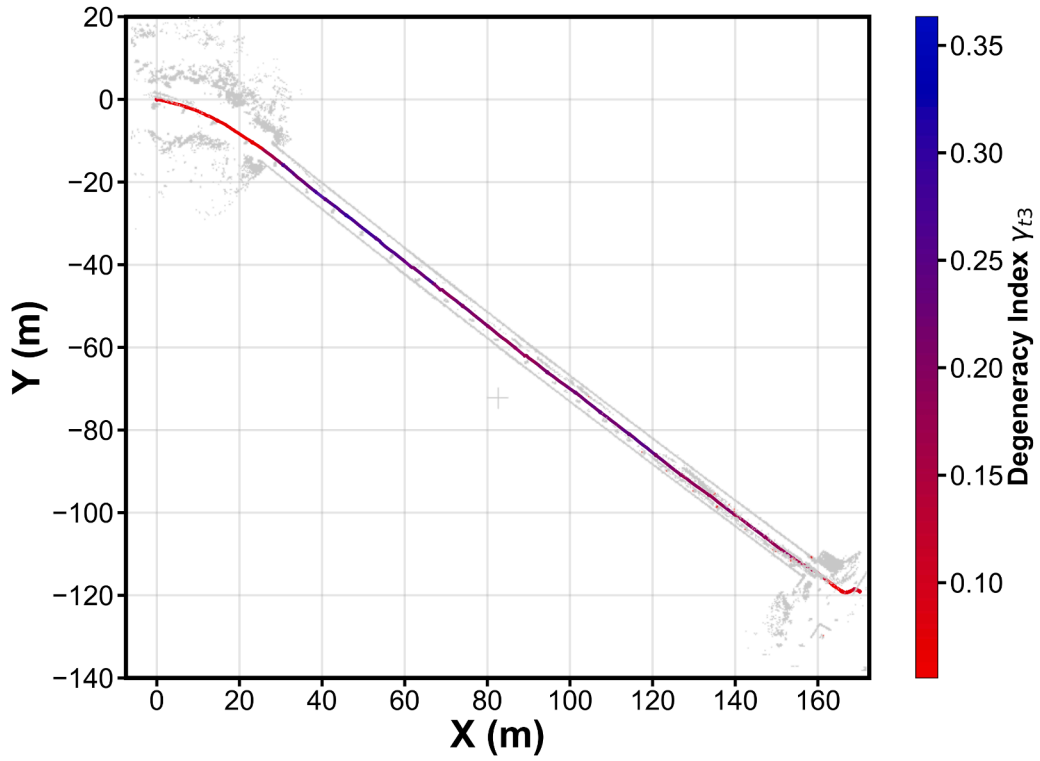


Fig. 8. Spatial distribution of γ_{t3} in Tunnel 3. The trajectory is color-coded based on the translation degeneracy index and superimposed on the gray-scale point cloud map. The colored trajectory indicates the variation in degeneracy index γ_{t3} along the trajectory, with blue denoting high values (severe degeneracy) and red denoting low values.

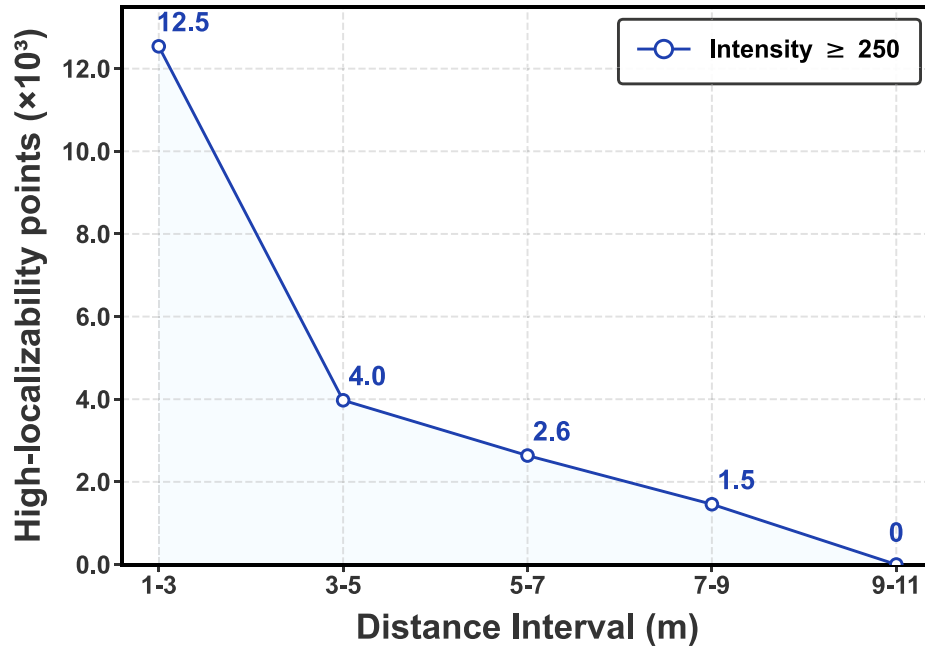


Fig. 9. Identification of high-localizability points as a function of inter-device distance.

in Tunnel 3, with the BR positioned at the tunnel's central axis while the MR executed a controlled linear displacement along the straight trajectory. Spatial data acquisition covered four discrete distance ranges: 1–3 m, 3–5 m, 5–7 m, and 7–9 m, and the result is shown in Fig. 9. Within 1–3 m, the system detects 12,500 points, progressively decreasing to 4000 (3–5 m), 2600 (5–7 m), and 1500 (7–9 m). No detectable points

meet the intensity threshold beyond 9 m. This consistent identification of at least 1500 high-intensity points across the 1–9 m range confirms their viability as localization constraints for tunnel SLAM.

This result validates the stable identifiability of the BR's beacons and offers an experimental foundation for defining the length of a basic motion unit of the leapfrogging advancement strategy detailed in

Table 2
Comparison of trajectory start-to-end error of different algorithms.

Dataset		Tunnel 1		Tunnel 2		Tunnel 3	
Ground Truth (m)		247.21		1002.90		207.96	
Algorithm		Est (m)	Err (m)	Est (m)	Err (m)	Est (m)	Err (m)
Multi	Leap-SLAM	246.81	-0.40	1003.06	+0.16	207.77	-0.19
	Zhang et al.'s [19]	231.78	-15.43	453.79	-549.11	94.11	-113.85
	Lamp2.0 [11]	235.82	-11.39	841.92	-160.98	74.52	-133.44
	Swarm-SLAM [41]	239.72	-7.49	935.02	-67.88	54.94	-153.02
	DiSCo-SLAM [18]	850.62	invalid	—	invalid	—	invalid
Single	Fast-LIO2 [9]	229.52	-17.69	339.68	-663.22	88.18	-119.78
	Fast-LIO2 / filtered	211.48	-35.73	515.41	-487.49	105.06	-102.90
	LIO-SAM [8]	—	invalid	—	invalid	—	invalid
	LIO-SAM / filtered	—	invalid	—	invalid	—	invalid
	D-LIO [30]	5165.18	invalid	50369.58	invalid	1076.44	invalid
	D-LIO / filtered	162.89	-84.32	182.70	-820.20	1406.52	invalid
	KISS-ICP [28]	103.72	-143.49	384.82	-618.08	132.77	-75.19
	KISS-ICP / filtered	137.43	-109.78	364.16	-638.74	110.88	-97.08
	Point-LIO [29]	42.80	-204.41	45.25	invalid	71.06	-136.90
	Point-LIO / filtered	47.97	-199.24	53.61	invalid	74.88	-133.08

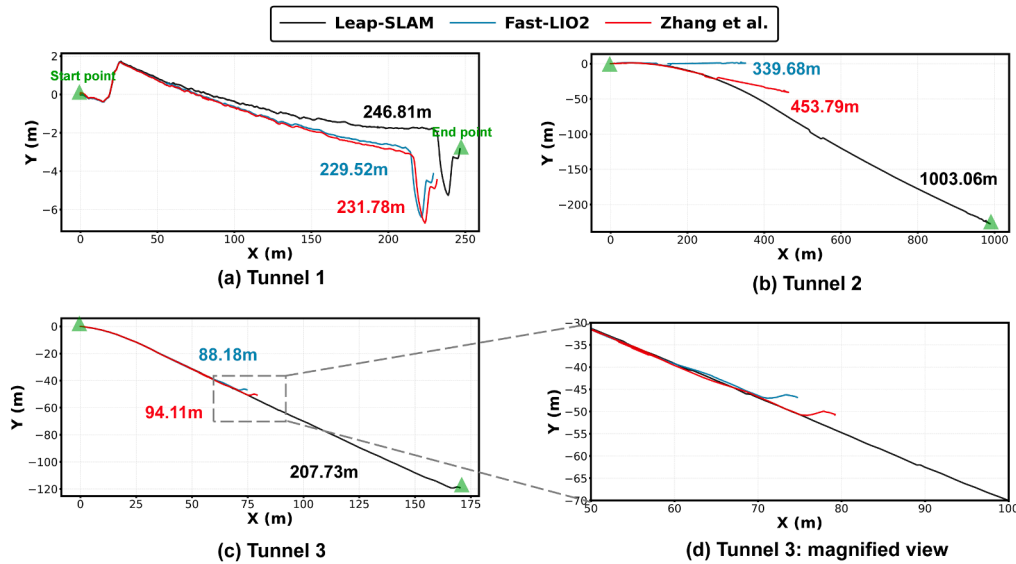


Fig. 10. The trajectory results of different algorithms on the three tunnel datasets. (a–c) present the estimated trajectories in Tunnel 1, Tunnel 2, and Tunnel 3, respectively. (d) provides a magnified view highlighting the ending points of Fast-LIO2 and Zhang et al.'s method.

Section 3.3.3. Given that the high-localizability points remain reliably detectable within 9 m under these experimental conditions, the maximum step length was constrained to 8 m to maintain a 1-meter safety margin. It should be noted that while the distance optimization experiments were primarily conducted in long, straight tunnels—where geometric degradation is most severe—the underlying methodology remains applicable to nonlinear scenarios. As established by the functional relationship between high-localizability features and inter-robot distance (Fig. 9), the system is designed to adaptively reduce the motion unit when encountering higher curvature. In such cases, although the effective visibility range may decrease, the determined step length ensures that the multi-robot system consistently observes the required features, thereby preventing sight loss and maintaining SLAM robustness.

4.3. Trajectory evaluation

To comprehensively validate the localization stability and accuracy of Leap-SLAM in degenerate tunnels, this section presents both qualitative and quantitative evaluations of the estimated trajectories. The proposed framework is implemented based on the Fast-LIO2 [9]

foundation. For a rigorous performance benchmark, we selected a diverse range of state-of-the-art (SOTA) baselines, encompassing representative single-robot LiDAR–inertial odometry approaches such as Fast-LIO2 [9], LIO-SAM [8], D-LIO [30], KISS-ICP [28], and Point-LIO [29], as well as advanced multi-robot SLAM systems including Lamp 2.0 [11], Swarm-SLAM [41], DiSCo-SLAM [18], and Zhang et al.'s method [19]. These particular algorithms were selected based on their technical representativeness across different SLAM paradigms (e.g., filter-based vs. optimization-based) and their specific environmental relevance to underground or degraded scenarios. In particular, Lamp 2.0 was included to evaluate the system's performance against established multi-robot frameworks designed for subterranean exploration.

Specifically, to ensure a fair comparison and investigate whether the BR's presence affects registration, each single-robot baseline was evaluated in two modes: the standard version using the full point cloud and a filtered version with the BR's points explicitly removed (denoted as “/ filtered”). Regarding multi-robot approaches, the systems operate in a full collaborative mode by processing data from both robots to calculate relative constraints. To ensure scientific rigor, we report only the trajec-

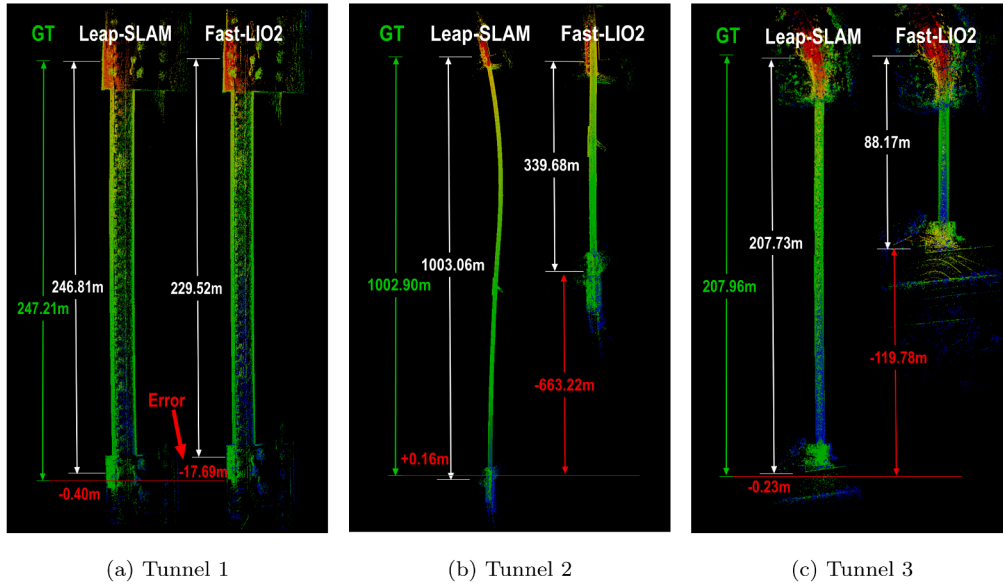


Fig. 11. Global point cloud maps of Leap-SLAM and Fast-LIO2 across three tunnel datasets. For each subfigure: ground truth length (green text), estimated length (white text), and absolute length error (red text).

tory of the Master Robot (MR) in our benchmarks, as high-precision RTK ground truth was exclusively available for the MR.

Due to GNSS signal blockage inside the tunnel, the RTK system cannot provide reliable positioning. Therefore, an open area outside the tunnel was selected as the starting point, where the RTK system recorded precise ground-truth coordinates. The system then traversed through the tunnel and exited at another open area, where the terminal position was also measured using RTK for reference. In this way, we can obtain the GT values of the start-to-end trajectories of the three tunnels through GNSS measurements, which are 247.21 m, 1002.90 m, and 207.96 m respectively. The start-to-end distance deviation between the estimated trajectory and the RTK reference trajectory was computed as the trajectory start-to-end error.

The experimental results summarized in Table 2 reveal a significant performance gap between the proposed Leap-SLAM and the comparative baselines in tunnel datasets. The analysis of these results can be categorized into three levels of performance:

First, classic frameworks such as LIO-SAM, D-LIO, and DiSCo-SLAM proved inadequate for these challenging datasets. These methods either terminated prematurely (denoted by “–” in Table 2) or produced estimations with extreme drift. In this study, results are formally classified as “invalid” if the start-to-end trajectory error exceeds the total ground truth length of the tunnel. For instance, D-LIO’s estimation in Tunnel 1 diverged to 5165.18 m—more than 20 times the 247.21 m ground truth. Consequently, these invalid methods were excluded from the trajectory visualization in Fig. 10 to maintain a legible coordinate field of view and prevent visualization distortion. In the figure, the start and end points of the RTK trajectories are marked with green triangles.

Second, a group of more recent methods, including Fast-LIO2, Lamp2.0, Swarm-SLAM, and Zhang et al.’s, managed to complete the datasets but suffered from substantial drift. In Tunnel 2 and Tunnel 3, their trajectory errors often exceeded 50% of the total tunnel length. Notably, the performance of their filtered versions (denoting the removal of BR point clouds from the MR raw data) did not show decisive improvements. This critical observation suggests that mutual interference between robots is a secondary concern; the primary bottleneck remains the acute geometric degeneracy inherent in tunnels, which these baselines fail to mitigate even when dynamic obstacles are removed.

In sharp contrast, the proposed Leap-SLAM demonstrated exceptional robustness and precision across all datasets. Most notably, in the

1002.90 m Tunnel 2, Leap-SLAM restricted the end-to-end error to a marginal 0.16 m, achieving a drift rate of approximately 0.016%. As illustrated in Fig. 10, while the trajectories of other methods span only a fraction of the actual distance due to severe degradation, the Leap-SLAM trajectory aligns consistently with the RTK ground truth, validating its superior consistency over long-range trajectories.

4.4. Mapping evaluation

Beyond trajectory assessment, the quality of map reconstruction is crucial, whose completeness and consistency not only reflect the effectiveness of spatial geometric constraint optimization within the system but also serve as key indicators for evaluating its capacity to suppress degeneracy. For mapping evaluation, only a representative algorithm (Fast-LIO2) and our own are selected for comparison, given the poor trajectory performance of the other algorithms previously discussed.

Fig. 11 presents global point cloud maps of Leap-SLAM and Fast-LIO2 across three tunnel datasets. From a qualitative structural perspective, Leap-SLAM’s global point cloud results for the three tunnels are largely consistent with the tunnel images in Fig. 6 and the long-branch structural design of the Google Maps route. Furthermore, Leap-SLAM’s point cloud mapping exhibits high completeness; the discrepancy between the point cloud’s end-to-end length (corresponding to the trajectory’s start and end points) and the ground truth values is only tens of centimeters, whereas Fast-LIO2’s point cloud mapping in Tunnel 2 and Tunnel 3 covers only about 50% of the designed structural length. This indicates that Leap-SLAM’s global point cloud mapping achieves far greater completeness than Fast-LIO2, attributable to the lack of drift-induced degeneracy during tunnel traversal.

In addition to global point cloud structure consistency, Fig. 12 compares the local point cloud map consistency between Leap-SLAM and Fast-LIO2 in Tunnel 1. Our approach entails continuously zooming in on local point cloud details to assess if prominent tunnel objects (e.g., stationary vehicles) exhibit ghosting, to identify potential drift or degeneracy. In the local detailed view, Leap-SLAM clearly reconstructs aligned vehicle outlines, while Fast-LIO2 exhibits blurry point cloud duplication. This demonstrates Leap-SLAM’s high consistency in refining local point cloud maps and compensating for local environmental degeneracy rather than Fast-LIO2.

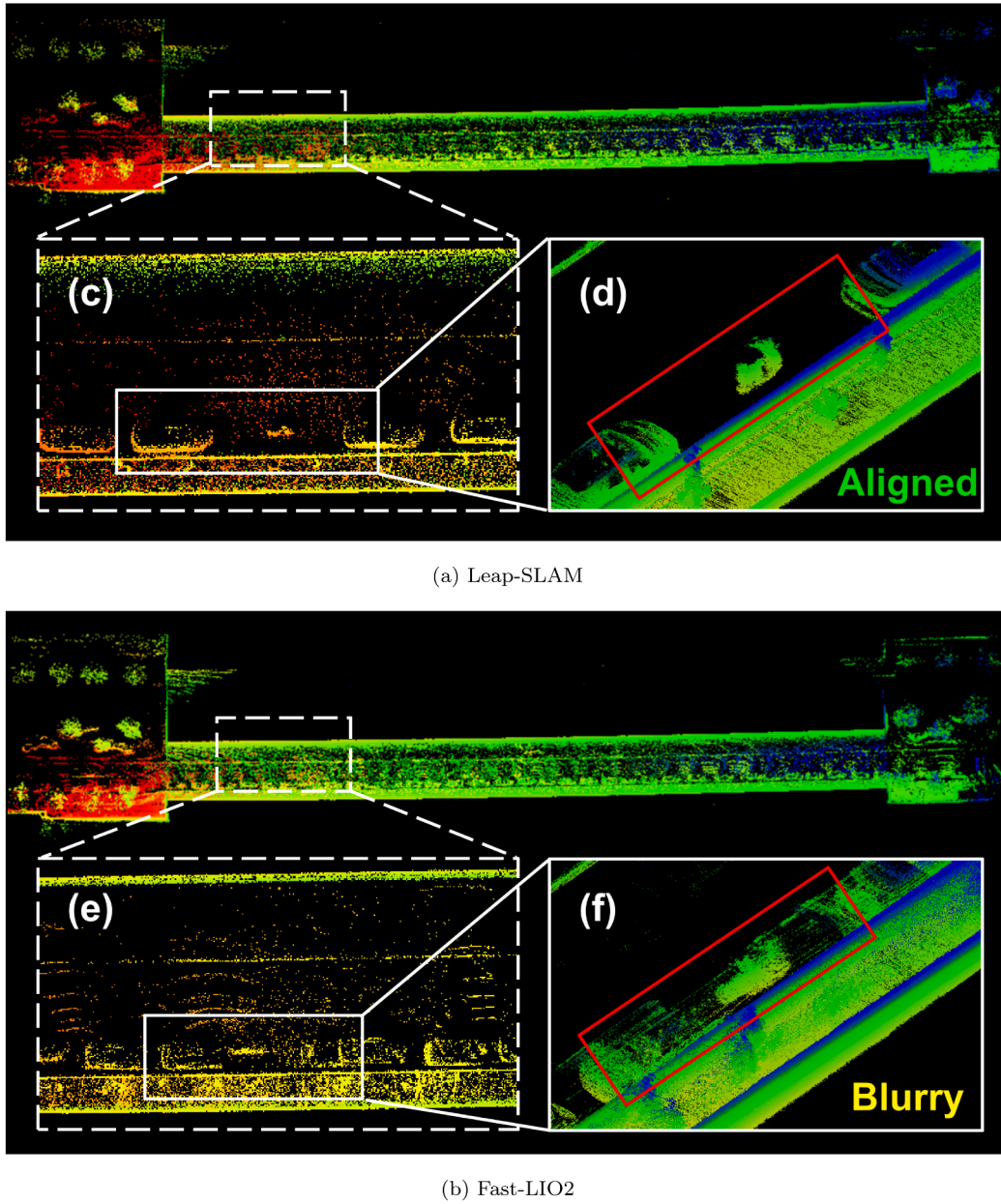


Fig. 12. Consistency comparison of local point cloud maps between Leap-SLAM and Fast-LIO2 in Tunnel 1. (c) and (e) are local views inside the tunnel, while (d) and (f) are more detailed views to examine the drift.

Due to the absence of a higher precision reference point cloud of the tunnels, the qualitative analysis of point cloud mapping was constrained. Therefore, we deployed several reflective markers inside the tunnel and accurately measured the line segment distances between the markers, thereby addressing the gaps in the quantitative evaluation of the point cloud. Fig. 13 shows reflective markers mounted on the tunnel walls. The reference lengths between consecutive markers were measured using a laser rangefinder. Each segment between two adjacent markers defines an evaluation node. Fig. 14 details the distance measurement for Segment 1 of Tunnel 1: two reflective markers were identified via their high-intensity points in the point cloud, and their geometric centers were selected to compute the inter-marker distance. The segment error (SE)—defined as the deviation between the SLAM mapping value and the reference value for each segment—was calculated. The Absolute Average Error (AAE)—defined as the mean of the multiple segment errors—was further introduced to characterize map accuracy

inside the tunnel. The five markers in Tunnel 1 form four segments when paired together, while the six markers in Tunnel 2 form five segments. Table 3 compares the SE and AAE for Leap-SLAM and Fast-LIO2, based on rangefinder-derived reference values. Leap-SLAM achieved an AAE of 0.13 m in Tunnel 1 (0.04 m in one segment) and 0.28 m in Tunnel 3. In contrast, Fast-LIO2 exhibited substantially higher errors in both tunnels—an AAE of 4.12 m in Tunnel 1 and 23.84 m in Tunnel 3. Critically, in Tunnel 3, every segment exhibited segment errors of more than 20 m, indicating complete failure in geometric constraint utility.

In summary, from global to local scales and qualitative to quantitative analyses, multiple approaches have consistently demonstrated Leap-SLAM's capability for robust and high-fidelity global and local mapping. Its ability to suppress drift in degraded tunnel environments further underscores the effectiveness of our proposed leapfrogging collaboration strategy and high-localizability feature constraint mechanism.

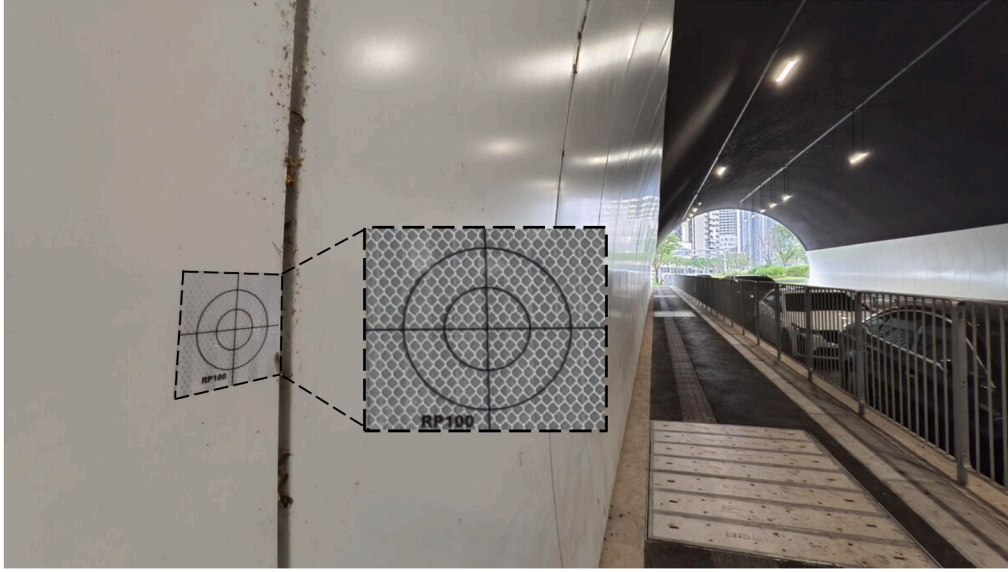


Fig. 13. Reflective marks installed in Tunnel 1 for quantitative analysis of point clouds inside tunnels.

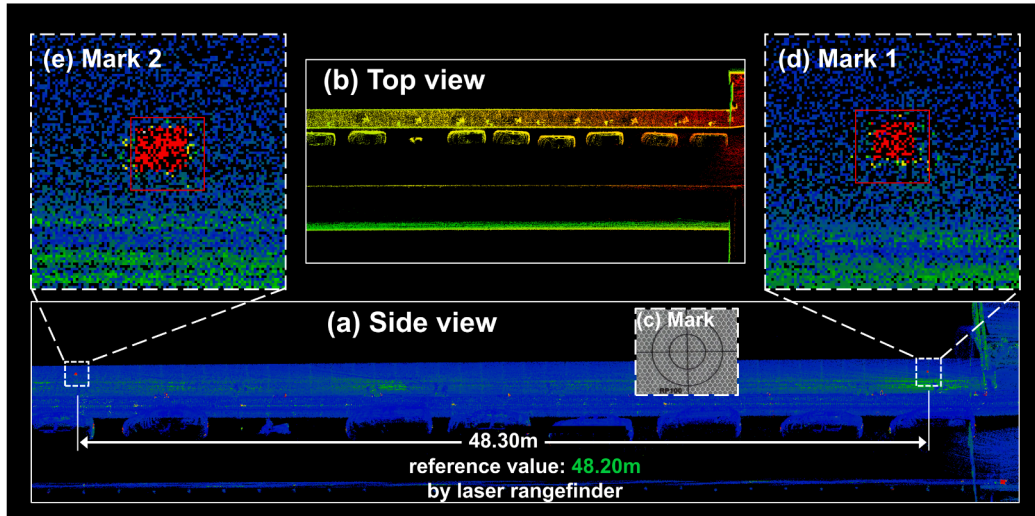


Fig. 14. Local point cloud mapping evaluation for Segment 1 of Tunnel 1. (a) and (b) represent the side view and top view of local point cloud mapping respectively. (d) and (e) are the magnified views of two reflective marks in (a), forming Segment 1. The reference value, measured with a laser rangefinder, is 48.20 m, while the SLAM-mapped value is 48.30m.

Table 3

SE and AAE for Leap-SLAM and Fast-LIO2 based on rangefinder-derived reference values (Unit: m).

Dataset	Type	Algorithm	Seg 1	Seg 2	Seg 3	Seg 4	Seg 5	Avg/AAE
Tunnel 1	Reference		48.20	48.20	48.20	55.43	–	50.01
	SE	Leap-SLAM	0.10	0.11	0.04	0.27	–	0.13
		Fast-LIO2	-8.65	-3.69	-3.62	-0.54	–	4.12
Tunnel 3	Reference		29.98	30.01	30.20	30.17	29.99	30.07
	SE	Leap-SLAM	-0.34	0.56	0.05	-0.38	-0.06	0.28
		Fast-LIO2	-20.08	-20.11	-25.80	-28.97	-24.22	23.84

4.5. Reason analysis of the degeneracy mitigation

This section analyzes how incorporating high-localizability constraints compensates for tunnel degeneracy and enhances trajectory and mapping performance. As previously analyzed, when the tunnel degeneracy index calculated from the original information matrix of environmental observations is high and exceeds the threshold range (see

in Section 3.2.2), SLAM systems incorporating environmental feature constraints will exhibit severe degeneracy. Therefore, we thus compare whether using a multi-robot leapfrog strategy to add high-localizability feature constraints reduces the degeneracy index exceeding the threshold range to a stable range.

Fig. 15 compares the variation of forward translation degeneracy indices γ_{13} in Tunnel 3 with and without high-localizability

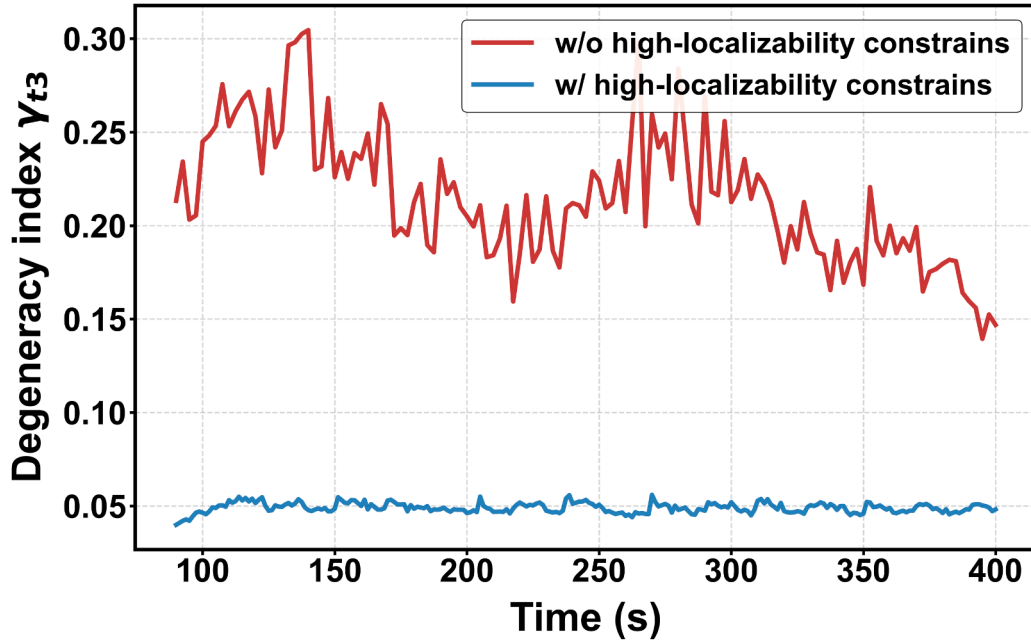


Fig. 15. Comparison of forward translation degeneracy indices γ_{t3} inside Tunnel 3 with and without high-localizability constraints.

Table 4
Computational performance and memory overhead of Leap-SLAM.

Metrics	Tunnel 1	Tunnel 2	Tunnel 3
Trajectory Length (m)	247.21	1002.9	207.96
<i>Time per Frame (ms):</i>			
Degeneracy Analysis	6.0	6.2	6.8
Feature Extraction	0.6	0.7	0.7
Degeneracy Mitigation	21.8	22.1	23.3
Total Avg. Time	28.4	29.0	30.8
<i>Memory Usage (MB):</i>			
Average RAM	359.41	426.65	360.54
Peak RAM	435.80	704.12	432.42

constraints. Without these constraints, the degeneracy index γ_{t3} remains high, surpassing 0.15 throughout the 90–400 s interval inside the tunnel, indicating weakened observability along the forward translation axis in tunnels. With these high-localizability constraints, the degeneracy index γ_{t3} is significantly reduced to around 0.05, a relatively low level. A low degeneracy index enables the information matrix to maintain well-behaved properties, facilitating the stable solution of the objective equation, thereby ensuring SLAM accuracy and stability [42–44].

This analysis explains that degeneracy mitigation is fundamentally driven by high-localizability feature constraints reducing the degeneracy index below the threshold, mathematically guaranteeing the stability of the state estimation objective equation. These findings validate the experimental efficacy and theoretical consistency of our degeneracy-mitigated robust SLAM with leapfrogging multi-robot collaboration approach in degraded tunnel environments.

4.6. Computational efficiency and resource utilization

To evaluate the potential for real-time deployment on resource-constrained platforms, we analyzed the computational efficiency of the proposed Leap-SLAM. The evaluation was conducted on a system equipped with an Intel i7-12700 CPU and 16 GB of RAM.

4.6.1. Per-frame processing time

Beyond total execution time, we monitored the runtime of three core modules for each LiDAR scan: degeneracy analysis, high-localizability feature extraction, and degeneracy mitigation. As summarized in Table 4, the total average processing runtime remains consistently between 28.4 ms and 30.8 ms across all datasets.

Table 4 reveals that degeneracy mitigation is the most computationally intensive component, accounting for the majority of the total per-frame runtime. In contrast, the degeneracy analysis and high-localizability feature extraction modules are highly efficient, with the latter requiring only 0.7 ms on average.

Even with these integrated modules, the total runtime (e.g., 29.0 ms in Tunnel 2) is significantly lower than the 100 ms threshold required for real-time operation with a standard 10 Hz LiDAR. This substantial computational margin—approximately 70% of the available time budget—provides a robust foundation for integrating auxiliary high-level tasks or deploying the system on platforms with more restricted hardware resources.

4.6.2. Memory footprint

In terms of memory utilization, the average consumption remains stable between 359.41 MB and 426.65 MB across the tested environments. Even in the extended Tunnel 2 mission, which spans over 1000 m, the peak memory usage is restricted to 704.12 MB. Given the 16 GB of RAM available on our experimental platform, the memory overhead of Leap-SLAM is remarkably low, leaving more than 95% of the system resources available for other concurrent processes. These results confirm that Leap-SLAM maintains a lightweight memory footprint, making it well-suited for long-term deployment on mobile robots with limited on-board memory resources.

5. Conclusion

In this work, we proposed Leap-SLAM, a multi-robot framework that enhances LiDAR-inertial odometry by fusing high-localizability constraints via a leapfrogging collaboration strategy. By deploying a beacon robot to actively generate high-localizability constraints, the system effectively mitigates structural degeneracy, particularly the scarcity of longitudinal features in tunnels. Experimental results across kilometer-scale tunnels demonstrate exceptional stability—achieving a 16 cm drift

over 1,000 m—and high-fidelity mapping, confirming its efficacy for autonomous monitoring in GNSS-denied, degraded environments.

Currently, the leapfrogging strategy remains relatively rigid; fixed motion steps may hinder visual contact in sharp curves, and the beacon robot lacks autonomous navigation and obstacle avoidance. Furthermore, the observation noise for high-localizability points is predefined as a constant. Future work will integrate embodied multi-modal large models and reinforcement learning to enable adaptive collaborative control. We aim to develop autonomous modules that dynamically adjust trajectories based on environmental curvature and sensor visibility, while incorporating distance-adaptive noise models to further enhance robustness in complex scenarios.

CRedit authorship contribution statement

Shoubin Chen: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization; **Jiasheng Chen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation; **Baiyang Zhang:** Writing – review & editing, Software, Methodology, Data curation; **Maosheng Yan:** Writing – review & editing, Validation, Formal analysis; **Jianping Li:** Writing – review & editing, Conceptualization; **Bo Zhang:** Writing – review & editing, Resources, Conceptualization; **Qingquan Li:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by **Shenzhen Science and Technology Program (QNXMB20250701100701002, ZDCY20250901104706008)** and **National Natural Science Foundation of China(42404046,62173234)**.

Data availability

Data will be made available on request.

References

- [1] H. Xu, M. Wang, C. Liu, Y. Guo, Z. Gao, C. Xie, Tunnel crack assessment using simultaneous localization and mapping (SLAM) and deep learning segmentation, *Autom. Constr.* 171 (2025) 105977.
- [2] Y. Jiang, L. Wang, B. Zhang, X. Dai, J. Ye, B. Sun, N. Liu, Z. Wang, Y. Zhao, Tunnel lining detection and retrofitting, *Autom. Constr.* 152 (2023) 104881.
- [3] J. Cheng, L. Zhang, Q. Chen, X. Hu, J. Cai, A review of visual SLAM methods for autonomous driving vehicles, *Eng. Appl. Artif. Intell.* 114 (2022) 104992.
- [4] S. Chen, Z. Wu, K. Zhang, C. Li, B. Zhang, F. Ma, F.R. Yu, Q. Li, Exploring embodied multimodal large models: development, datasets, and future directions, *Inf. Fusion* 122 (2025) 103198.
- [5] R. Mur-Artal, J.M.M. Montiel, J.D. Tardos, ORB-SLAM: A versatile and accurate monocular SLAM system, *IEEE Trans. Robot.* 31 (5) (2015) 1147–1163.
- [6] J. Zhang, S. Singh, et al., LOAM: Lidar odometry and mapping in real-time, in: *Robotics: Science and Systems*, Vol. 2, Berkeley, CA, 2014, pp. 1–9.
- [7] T. Shan, B. Englot, Lego-Loam: lightweight and ground-optimized lidar odometry and mapping on variable terrain, in: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, 2018, pp. 4758–4765.
- [8] T. Shan, B. Englot, D. Meyers, W. Wang, C. Ratti, D. Rus, Lio-sam: tightly-coupled lidar inertial odometry via smoothing and mapping, in: 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, 2020, pp. 5135–5142.
- [9] W. Xu, Y. Cai, D. He, J. Lin, F. Zhang, Fast-lío2: fast direct lidar-inertial odometry, *IEEE Trans. Robot.* 38 (4) (2022) 2053–2073.
- [10] Y. Cong, C. Chen, B. Yang, J. Li, W. Wu, Y. Li, Y. Yang, 3D-CSTM: a 3D continuous spatio-temporal mapping method, *ISPRS J. Photogramm. Remote Sens.* 186 (2022) 232–245.
- [11] Y. Chang, K. Ebadi, C.E. Denniston, M.F. Ginting, A. Rosinol, A. Reinke, M. Palieri, J. Shi, A. Chatterjee, B. Morrell, et al., LAMP 2.0: a robust multi-robot SLAM system for operation in challenging large-scale underground environments, *IEEE Robot. Autom. Lett.* 7 (4) (2022) 9175–9182.
- [12] J. Zhang, M. Kaess, S. Singh, On degeneracy of optimization-based state estimation problems, in: 2016 IEEE International Conference on Robotics and Automation (ICRA), IEEE, 2016, pp. 809–816.
- [13] A. Hinduja, B.-J. Ho, M. Kaess, Degeneracy-aware factors with applications to underwater slam, in: 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, 2019, pp. 1293–1299.
- [14] T. Tuna, J. Nubert, Y. Nava, S. Khattak, M. Hutter, X-icp: localizability-aware lidar registration for robust localization in extreme environments, *IEEE Trans. Robot.* 40 (2023) 452–471.
- [15] F. Han, H. Zheng, W. Huang, R. Xiong, Y. Wang, Y. Jiao, DAMS-LIO: a degeneracy-aware and modular sensor-fusion LiDAR-inertial odometry, 2023, arXiv:2302.01703.
- [16] S. Saedi, M. Trentini, M. Seto, H. Li, Multiple-robot simultaneous localization and mapping: a review, *J. Field Robot.* 33 (1) (2016) 3–46.
- [17] M. Kegeles, G. Grisetti, M. Birattari, Swarm slam: challenges and perspectives, *Front. Robot. AI* 8 (2021) 618268.
- [18] Y. Huang, T. Shan, F. Chen, B. Englot, Disco-SLAM: distributed scan context-enabled multi-robot lidar slam with two-stage global-local graph optimization, *IEEE Robot. Autom. Lett.* 7 (2) (2021) 1150–1157.
- [19] B. Zhang, Z. Xiong, J. Qiu, S. Chen, Y. Hu, S. Chen, Distributed robust communication-efficient multi-robot SLAM combining real-time intersection and historical loop constraints, *IEEE Internet Things J.* 12, 24893–24903 (2025).
- [20] Y. Zhuang, X. Sun, Y. Li, J. Huai, L. Hua, X. Yang, X. Cao, P. Zhang, Y. Cao, L. Qi, et al., Multi-sensor integrated navigation/positioning systems using data fusion: from analytics-based to learning-based approaches, *Inf. Fusion* 95 (2023) 62–90.
- [21] P.J. Besl, N.D. McKay, Method for registration of 3-D shapes, in: *Sensor Fusion IV: Control Paradigms and Data Structures*, 1611, SPIE, 1992, pp. 586–606.
- [22] P. Biber, W. Straßer, The normal distributions transform: a new approach to laser scan matching, in: *Proceedings 2003 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2003)*(Cat. No. 03CH37453), Vol. 3, IEEE, 2003, pp. 2743–2748.
- [23] P. Dellenbach, J.-E. Deschaud, B. Jacquet, F. Goulette, Ct-icp: real-time elastic lidar odometry with loop closure, in: 2022 International Conference on Robotics and Automation (ICRA), IEEE, 2022, pp. 5580–5586.
- [24] E. Einhorn, H.-M. Gross, Generic NDT mapping in dynamic environments and its application for lifelong SLAM, *Robot. Autom. Syst.* 69 (2015) 28–39.
- [25] X. Duan, Q. Hu, M. Ai, P. Zhao, M. Wu, J. Li, C. Xiong, STATIC-LIO: s sliding window and terrain-assisted dynamic points removal LiDAR inertial odometry, *Inf. Fusion* 121 (2025) 103132. <https://doi.org/10.1016/j.inffus.2025.103132>
- [26] W. Xu, F. Zhang, Fast-lío: a fast, robust lidar-inertial odometry package by tightly-coupled iterated kalman filter, *IEEE Robot. Autom. Lett.* 6 (2) (2021) 3317–3324.
- [27] D. He, W. Xu, F. Zhang, Embedding manifold structures into Kalman filters, 2021, arXiv:2102.03804, <https://api.semanticscholar.org/CorpusID:231847324>.
- [28] I. Vizzo, T. Guadagnino, B. Mersch, L. Wiesmann, J. Behley, C. Stachniss, Kiss-icp: in defense of point-to-point icp—simple, accurate, and robust registration if done the right way, *IEEE Robot. Autom. Lett.* 8 (2) (2023) 1029–1036.
- [29] D. He, W. Xu, N. Chen, F. Kong, C. Yuan, F. Zhang, Point-LIO: robust high-bandwidth light detection and ranging inertial odometry, *Adv. Intell. Syst.* 5 (7) (2023) 2200459.
- [30] H. Devarajan, H. Zheng, A. Kougkas, X.-H. Sun, V. Vishwanath, Dlio: a data-centric benchmark for scientific deep learning applications, in: 2021 IEEE/ACM 21st International Symposium on Cluster, Cloud and Internet Computing (CCGrid), IEEE, 2021, pp. 81–91.
- [31] Y. Cong, C. Chen, B. Yang, R. Zhong, S. Sun, Y. Xu, Z. Yan, X. Zou, Z. Tu, OR-LIM: Observability-aware robust LiDAR-inertial-mapping under high dynamic sensor motion, *ISPRS J. Photogramm. Remote Sens.* 218 (2024) 610–627.
- [32] J. Zhang, H. Zhang, X. Zhou, B. Chen, X. Wang, T. Xu, A graph-based SLAM method assisted by visual marker in the degenerate scenes, in: 2023 International Conference on Advanced Robotics and Mechatronics (ICARM), IEEE, 2023, pp. 1–6.
- [33] J. Hatleskog, K. Alexis, Probabilistic degeneracy detection for point-to-plane error minimization, *IEEE Robot. Autom. Lett.* 9 (2024) 11234–11241.
- [34] W. Chen, X. Wang, S. Gao, G. Shang, C. Zhou, Z. Li, C. Xu, K. Hu, Overview of multi-robot collaborative SLAM from the perspective of data fusion, *Machines* 11 (6) (2023) 653.
- [35] X. Liu, S. Wen, J. Zhao, T.Z. Qiu, H. Zhang, Edge-assisted multi-robot visual-inertial SLAM with efficient communication, *IEEE Trans. Autom. Sci. Eng.* 22 (2024) 2186–2198.
- [36] J. Kim, H. Kim, S. Jeong, Y. Shin, Y. Cho, Diter ++: diverse terrain and multimodal dataset for multi-robot slam in multi-session environments, in: 2025 IEEE International Conference on Robotics and Automation (ICRA), IEEE, 2025, pp. 12187–12193.
- [37] K. Ebadi, Y. Chang, M. Palieri, A. Stephens, A. Hatteland, E. Heiden, A. Thakur, N. Funabiki, B. Morrell, S. Wood, et al., LAMP: Large-scale autonomous mapping and positioning for exploration of perceptually-degraded subterranean environments, in: 2020 IEEE International Conference on Robotics and Automation (ICRA), IEEE, 2020, pp. 80–86.
- [38] Z. Liao, X. Zhang, T. Zhang, Z. Li, Z. Zheng, Z. Wen, Y. Li, A real-time degeneracy sensing and compensation method for enhanced LiDAR SLAM, *IEEE Trans. Intell. Transp. Syst.* 26 (2025) 4202–4213.
- [39] C.H. Tong, P. Furgale, T.D. Barfoot, Gaussian process Gauss-Newton for non-parametric simultaneous localization and mapping, *Int. J. Robot. Res.* 32 (5) (2013) 507–525.

- [40] C. Rösmann, F. Hoffmann, T. Bertram, Integrated online trajectory planning and optimization in distinctive topologies, *Robot. Auton. Syst.* 88 (2017) 142–153.
- [41] P.-Y. Lajoie, G. Beltrame, Swarm-SLAM: sparse decentralized collaborative simultaneous localization and mapping framework for multi-robot systems, *IEEE Robot. Autom. Lett.* 9 (1) (2023) 475–482.
- [42] S. Boyd, L. Vandenberghe, *Convex Optimization*, vol. 25, Cambridge University Press, Cambridge, England, 2010.
- [43] S. Thrun, Probabilistic robotics, *Commun. ACM* 45 (3) (2002) 52–57.
- [44] M. Kaess, A. Ranganathan, F. Dellaert, iSAM: incremental smoothing and mapping, *IEEE Trans. Robot.* 24 (6) (2008) 1365–1378.